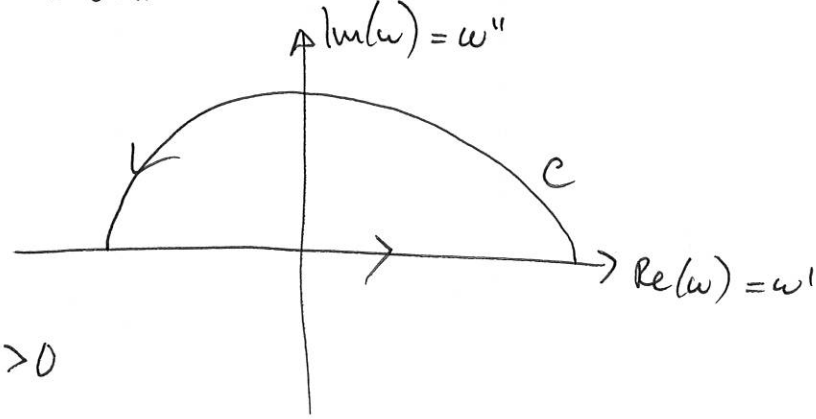
We know that $\chi(t) = -i\theta(t)\langle[\hat{A}(t), \hat{O}(0)]\rangle$ is retarded, i.e. $\chi(t)=0$ for $t<0$.

This leads to $\chi(\omega) = \int_0^\infty dt\, e^{i(\omega+i\delta)t}\,\chi(\omega)$ with $\delta=+0$.

Reversly, $\chi(t) = \int_{-\infty}^{\infty}\frac{d\omega}{2\pi}\, e^{-i\omega t}\,\chi(\omega)$

For $t<0$ we can close the integral in upper half-plane $\mathrm{Im}(\omega)>0$ of <u>complex</u> ω.



(There $e^{-i\omega t} = e^{-i(\omega'+i\omega'')t} = e^{-i\omega' t}\, e^{\omega'' t}$ converges in $\omega''\to\infty$ limit due to $t<0$.)

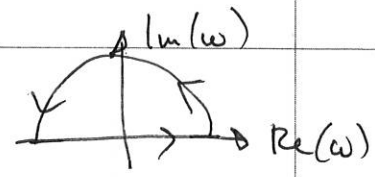
But $\chi(t<0)=0 \Rightarrow \int_C \frac{d\omega}{2\pi}\, e^{-i\omega t}\,\chi(\omega) = 0$

$\Rightarrow \chi(\omega)$ has no singularities and poles in semi-plane $\omega''=\mathrm{Im}(\omega)>0$.

Then, by Cauchy's integral formula,

$$\chi(z) = \frac{1}{2\pi i}\int_C \frac{\chi(\omega)}{\omega - z}\, d\omega$$

where C = any closed contour within and on which $\chi(z)$ is regular and z is any point inside C.

So let C be the countour above 

Since $\chi(\omega)$ has no poles in $\mathrm{Im}(\omega)>0$ region, and

$\chi(\omega) \to 0$ for $|\omega| \to \infty$ (which is physically natural condition)

$$\int_C d\omega \frac{\chi(\omega)}{-z+\omega} = \underbrace{\int d\omega \frac{\chi(\omega)}{-z+\omega}}_{\text{Large semi-circle}} + \underbrace{\int_{-\infty}^{+\infty} d\omega \frac{\chi(\omega)}{-z+\omega}}_{\text{(real axis)}}$$

Integral over the semi-circle vanishes $\Rightarrow$

$$2\pi i\, \chi(z) = \int_{-\infty}^{+\infty} d\omega \frac{\chi(\omega)}{\omega - z} \qquad (\mathrm{Im}\, z > 0)$$

Function on the real axis under the integral is understood as $\lim_{\varepsilon \to +0} \chi(x+i\varepsilon)$ where $\underline{x = \text{Real}}$.

let $z = x' + i\varepsilon$ too:

$$2\pi i\, \chi(x') = \int_{-\infty}^{\infty} dx \frac{\chi(x)}{x-(x'+i\varepsilon)} = \left(\frac{1}{x-x'-i\varepsilon} = \underbrace{P\frac{1}{x-x'}}_{\text{principal value}} + i\pi\delta(x-x') \right)$$

$$= P\int_{-\infty}^{\infty} dx \frac{\chi(x)}{x-x'} + i\pi\, \chi(x').$$

$$\Rightarrow \quad i\pi\, \chi(x') = P\int_{-\infty}^{\infty} dx \frac{\chi(x)}{x-x'}$$

$$\chi(x) = \mathrm{Re}\,\chi(x) + i\,\mathrm{Im}\,\chi(x) = \chi'(x) + i\chi''(x)$$

$x' \to y$ for convenience

$$i\pi\left(\chi'(y) + i\chi''(y)\right) = P\int_{-\infty}^{\infty} dx \frac{\chi'(x) + i\chi''(x)}{x-y}$$

And now we separate real and Im parts:

Kramers Kronig relations
$$\begin{cases} \chi'(y) = \frac{1}{\pi} P \int_{-\infty}^{\infty} dx \, \frac{\chi''(x)}{x-y} \\ \chi''(y) = -\frac{1}{\pi} P \int_{-\infty}^{\infty} dx \, \frac{\chi'(x)}{x-y} \end{cases}$$

Knowing $\mathrm{Im}\,\chi = \chi''$ we can find $\mathrm{Re}\,\chi = \chi'$!

For the electron gas $\chi''(\vec{q},\omega) = -N_0 \left(\frac{\pi\omega}{2v_F q} \theta\left(1 - \left|\frac{\omega}{v_F q}\right|\right)\right)$

Then $\chi'(\vec{q},\Omega) = \frac{1}{\pi} P \int_{-\infty}^{\infty} d\omega \, \frac{\chi''(\vec{q},\omega)}{\omega - \Omega} =$

$$= \frac{1}{\pi}\left(-\frac{N_0 \pi}{2v_F q}\right) \int_{-\infty}^{\infty} d\omega \, \frac{\omega \, \theta\left(1 - \frac{|\omega|}{v_F q}\right)}{\omega - \Omega} =$$

$$= \left(-\frac{N_0}{2v_F q}\right) \int_{-v_F q}^{v_F q} d\omega \, \frac{\omega}{\omega - \Omega} = \left(\frac{-N_0}{2v_F q}\right) \int_{-v_F q}^{v_F q} d\omega \, \frac{\omega - \Omega + \Omega}{\omega - \Omega} =$$

$$= \left(-\frac{N_0}{2v_F q}\right) \left(\omega + \Omega \ln(\omega - \Omega)\right)\Big|_{-v_F q}^{v_F q} =$$

$$= \left(-\frac{N_0}{2v_F q}\right) \left(2v_F q + \Omega \ln\left(\frac{v_F q - \Omega}{-v_F q - \Omega}\right)\right) =$$

$$= -N_0 \left\{ 1 + \frac{\Omega}{2v_F q} \ln \frac{\Omega - v_F q}{\Omega + v_F q} \right\} =$$

$$= N_0 \left\{ -1 + \frac{\Omega}{2v_F q} \ln \left| \frac{\Omega + v_F q}{\Omega - v_F q} \right| \right\}$$

which is exactly the result we had before!

Joule heating and $\mathrm{Im}(\chi)$

Recall how retarded susceptibility appears

$$\underbrace{\langle \hat{A}(\vec{r}) \rangle_t}_{\text{expectation value of } \hat{A}} = -\frac{i}{\hbar} \int_{-\infty}^{t} dt' \int d\vec{r}' \, \langle [\hat{A}(r,t), \hat{O}(r',t')] \rangle_0 \, h(\vec{r}',t')$$

in response to t-dependent perturbation $H'(t) = \int d\vec{r} \, h(\vec{r},t) \, \hat{O}(r)$

For the density-density susceptibility

$$\chi_0(\vec{r},t) = -i\,\theta(t) \langle [\hat{\rho}(r,t), \hat{\rho}(0)] \rangle$$

we have $\hat{A} = \hat{O} = \hat{\rho}$.

Perturbation $H'(t) = \int d\vec{r} \, \hat{\rho}(r,t) \underbrace{\varphi(r,t)}_{\text{(electrical) potential}}$

That creates (longitudinal) electric field

$$\vec{E}(r,t) = -\vec{\nabla} \varphi(r,t) \Rightarrow \vec{E}(\vec{q},\omega) = -i\,\vec{q}\,\varphi(\vec{q},\omega)$$

and electrical current $\vec{J}(r,t) = \sigma \vec{E}$ ← relation maybe nonlocal in $\vec{r}$-space but

$$\vec{J}(\vec{q},\omega) = \sigma(\vec{q},\omega) \, \vec{E}(\vec{q},\omega) \quad \text{(Ohm's law)}$$

We now find relation between σ and χ:

from continuity equation we know that

$$\partial_t \rho + \vec{\nabla} \cdot \vec{J} = 0 \Rightarrow -i\omega \rho(\vec{q},\omega) + i\,\vec{q} \cdot \vec{J}(\vec{q},\omega) = 0$$

$$\Rightarrow \vec{q}\cdot\vec{J}(\vec{q},\omega) = \sigma(\vec{q},\omega)\,\vec{q}\cdot\vec{E}(\vec{q},\omega) =$$

$$= \omega\,\rho(\vec{q},\omega) = \omega\left\{\chi(\vec{q},\omega)\,\varphi(\vec{q},\omega)\right\}$$

density appears in response to φ

But $\vec{q}\cdot\vec{E}(\vec{q},\omega) = \vec{q}\cdot\left(-i\vec{q}\,\varphi(\vec{q},\omega)\right) = -i\,q^2\,\varphi(\vec{q},\omega)$

$$\Rightarrow -i\,\sigma(\vec{q},\omega)\,q^2\,\varphi(\vec{q},\omega) = \omega\,\chi(\vec{q},\omega)\,\varphi(\vec{q},\omega)$$

Longitudinal conductivity $\quad \sigma(\vec{q},\omega) = \dfrac{i\omega}{q^2}\,\chi(\vec{q},\omega) \quad$ density-density susceptibility

$$\Rightarrow \mathrm{Re}\left[\sigma(\vec{q},\omega)\right] = -\frac{\omega}{q^2}\,\mathrm{Im}\left[\chi(\vec{q},\omega)\right]$$

$$\left(\text{because} \quad \mathrm{Re}\left(\frac{i\omega}{q^2}\chi(\vec{q},\omega)\right) = \mathrm{Re}\left(\frac{i\omega}{q^2}(\chi' + i\chi'')\right)\right).$$

We found previously (see posted lecture notes) that generally the rate of energy absorption is given by

$$\overline{\frac{dE}{dt}} = -2\,\omega\,\mathrm{Im}\left(\chi(\vec{q},\omega)\right)\,|\varphi(\vec{q},\omega)|^2 =$$

$$= 2\,q^2\,\mathrm{Re}\left(\sigma(\vec{q},\omega)\right)\,\underbrace{\varphi^*(\vec{q},\omega)}_{\left(\frac{\vec{E}(\vec{q},\omega)}{-i\vec{q}}\right)^*}\cdot\varphi(\vec{q},\omega)\qquad \cdot\,\frac{\vec{E}(\vec{q},\omega)}{-i\vec{q}}$$

$$= 2\,\mathrm{Re}\left[\sigma(\vec{q},\omega)\right]\,\left|\vec{E}(\vec{q},\omega)\right|^2$$

This is just Ohmic heat!

So we apply monochromatic $\vec{E}(t) = \vec{E}_0(q,\omega)\, e^{i(\vec{q}\cdot\vec{r}-\omega t)} + c.c.$
(complex conjugate)

It ~~also~~ drives current $\vec{J}(r,t) = \vec{J}_0(q,\omega)\, e^{i(\vec{q}\cdot\vec{r}-\omega t)} + c.c.$

Power dissipated at point $\vec{r}$, time t is

$$P(\vec{r},t) = \vec{J}\cdot\vec{E} = \left(\vec{J}_0(\bar{q},\omega)\, e^{i(\bar{q}\cdot\bar{r}-\omega t)} + \vec{J}_0^*(\bar{q},\omega)\, e^{-i(\bar{q}\cdot\bar{r}-\omega t)}\right)$$
$$\cdot\left(\vec{E}_0(\bar{q},\omega)\, e^{i(\bar{q}\cdot\bar{r}-\omega t)} + \vec{E}_0^*(\bar{q},\omega)\, e^{-i(\bar{q}\cdot\bar{r}-\omega t)}\right)$$

$$= \vec{J}_0(\bar{q},\omega)\cdot\vec{E}_0^*(\bar{q},\omega) + \vec{J}_0^*(\bar{q},\omega)\cdot\vec{E}_0(\bar{q},\omega)$$
$$+ \vec{J}_0\cdot\vec{E}_0\, e^{i2(\bar{q}\cdot\bar{r}-\omega t)} + \vec{J}_0^*\cdot\vec{E}_0^*\, e^{-i2(\bar{q}\cdot\bar{r}-\omega t)}$$

reversible terms
(describe energy going <u>in</u> and <u>out</u>)

absorption $$\overline{\frac{dE(\vec{r})}{dt}} = \frac{1}{T}\int_0^{T=2\pi/\omega} dt\, P(\vec{r},t) \Rightarrow \int_0^{2\pi/\omega} dt\, e^{\pm i2\omega t} =$$

$$= \frac{e^{\pm i2\omega t}\Big|_0^{2\pi/\omega}}{\pm i2\omega} = \frac{e^{i4\pi}-1}{\pm i2\omega} = 0.$$

$$\Rightarrow \overline{\frac{dE}{dt}} = \vec{J}_0(\bar{q},\omega)\cdot\vec{E}_0^*(\bar{q},\omega) + \vec{J}_0^*(\bar{q},\omega)\cdot\vec{E}_0(\bar{q},\omega)$$

By Ohm's law $\vec{J}_0(\bar{q},\omega) = \sigma(\bar{q},\omega)\,\vec{E}_0(\bar{q},\omega)$

$$\vec{J}_0^*(\bar{q},\omega) = \sigma^*(\bar{q},\omega)\,\vec{E}_0^*(\bar{q},\omega)$$

$$\Rightarrow \overline{\frac{dE}{dt}} = \sigma(\bar{q},\omega)\,\vec{E}_0(\bar{q},\omega)\cdot\vec{E}_0^*(\bar{q},\omega) + \sigma^*(\bar{q},\omega)\,|\vec{E}_0|^2 =$$
$$= 2\,\mathrm{Re}[\sigma(\bar{q},\omega)]\,|\vec{E}_0(\bar{q},\omega)|^2.$$

OK